Project: GridInsight - Smart Grid Monitoring & Renewable Energy Analytics System

1. Introduction

GridInsight is a web-based system designed for modern energy and utilities organizations to monitor grid health, track renewable energy generation, analyze load patterns, and support data-driven grid optimization. The system focuses on **smart grid visibility**, renewable asset performance, demand-response readiness, and sustainability reporting. It complements core utility systems by providing operational intelligence and analytics-driven decision support. It uses a **REST API-based backend architecture** and **Angular/React** for the frontend.

Compatible with **Java (Spring Boot)** frameworks and can also be adapted to **.NET (ASP.NET Core)**.

Actors / Users

- **Grid Operations Analyst**: Monitors grid KPIs, load patterns, and generation vs demand.
- **Renewable Asset Manager**: Tracks performance of solar/wind assets and generation forecasts.
- **Energy Planner**: Analyzes historical trends and prepares capacity and demand plans.
- **Sustainability & ESG Analyst**: Generates renewable contribution and carbon offset reports.
- **Utilities Admin**: Configures grid zones, asset groups, thresholds, users, and reports.

2. Module Overview

2.1 Identity & Access Management
Handles authentication, RBAC, and audit trails.

2.2 Grid Topology & Measurement Points
Defines grid zones, substations, feeders, and measurement points.

2.3 Renewable Generation Monitoring
Tracks renewable assets, generation data, and availability.

2.4 Load Monitoring & Demand Analysis
Captures consumption patterns and peak/off-peak metrics.

2.5 Forecasting & Grid Planning
Provides generation/load forecasts and capacity planning insights.

2.6 Sustainability & ESG Reporting
Generates renewable mix, emissions avoidance, and sustainability dashboards.

2.7 Alerts & Threshold Management

Raises in-app alerts for anomalies, overload risks, and generation drops.

3. Architecture Overview

- **Frontend**: Angular or React
- **Backend**: REST API-based architecture
- **Database**: Relational Database (MySQL/PostgreSQL/SQL Server)

4. Module-Wise Design

4.1 Identity & Access Management Module

Features:

- Register analyst/planner/asset manager/admin
- Login, RBAC, session management

Entities:

- **User**
  - UserID
  - Name
  - Role (GridAnalyst/AssetManager/Planner/ESG/Admin)
  - Email
  - Phone
- **AuditLog**
  - AuditID
  - UserID
  - Action
  - Resource
  - Timestamp
  - Metadata

4.2 Grid Topology & Measurement Points Module

Features:

- Define grid zones and measurement hierarchy

Entities:

- **GridZone**
  - ZoneID
  - Name
  - Region
  - VoltageLevel
  - Status (Active/Inactive)
- **MeasurementPoint**
  - PointID
  - ZoneID
  - AssetType (Substation/Feeder/Transformer)
  - Identifier
  - Unit (MW/MWh)
  - Status (Active/Inactive)

4.3 Renewable Generation Monitoring Module

Features:

- Monitor renewable asset output and availability

Entities:

- **RenewableAsset**
  - AssetID
  - AssetType (Solar/Wind/Hydro/Biomass)
  - Location
  - InstalledCapacityMW
  - CommissionDate
  - Status (Operational/UnderMaintenance/Offline)
- **GenerationRecord**
  - RecordID
  - AssetID
  - Timestamp
  - GeneratedEnergyMWh

    - AvailabilityPct

4.4 Load Monitoring & Demand Analysis Module

Features:

- Track demand, peaks, and load factors

Entities:

- **LoadRecord**
    - LoadID
    - ZoneID
    - Timestamp
    - DemandMW
    - DemandType (Actual/Estimated)
- **PeakEvent**
    - PeakID
    - ZoneID
    - StartTime
    - EndTime
    - PeakMW
    - Severity (Low/Medium/High)

4.5 Forecasting & Grid Planning Module

Features:

- Short-term and long-term demand and generation forecasting

Entities:

- **Forecast**
    - ForecastID
    - Scope (Zone/Asset/Period)
    - ForecastType (Generation/Load)
    - PeriodStart
    - PeriodEnd
    - ForecastValueMW

- ModelVersion
- **CapacityPlan**
  - PlanID
  - ZoneID
  - PlanningHorizon (Year/Quarter)
  - RecommendedCapacityMW
  - Notes

4.6 Sustainability & ESG Reporting Module

Features:

- Renewable contribution, carbon offset, ESG compliance reporting

Entities:

- **SustainabilityMetric**
  - MetricID
  - Period
  - RenewableSharePct
  - EmissionsAvoidedTons
  - GeneratedDate
- **ESGReport**
  - ReportID
  - ReportingStandard (GRI/CDP/Internal)
  - Period
  - GeneratedDate
  - Status (Draft/Published)

4.7 Alerts & Threshold Management Module

Features:

- Configurable thresholds and anomaly alerts

Entities:

- **ThresholdRule**
  - RuleID

    - Scope (Zone/Asset/Metric)
    - ThresholdValue
    - Comparison (Greater/Less)
- **Alert**
    - AlertID
    - RuleID
    - TriggeredAt
    - ActualValue
    - Severity (Low/Medium/High)
    - Status (Open/Acknowledged/Closed)

5. Deployment Strategy

- **Local**: Angular/React frontend, Spring Boot/.NET Core backend, local DB.
- **Production**: Cloud or on-prem deployment with secure APIs (self-contained, no external integrations).

6. Database Design

Tables:

- User(UserID, Name, Role, Email, Phone)
- AuditLog(AuditID, UserID, Action, Resource, Timestamp, Metadata)
- GridZone(ZoneID, Name, Region, VoltageLevel, Status)
- MeasurementPoint(PointID, ZoneID, AssetType, Identifier, Unit, Status)
- RenewableAsset(AssetID, AssetType, Location, InstalledCapacityMW, CommissionDate, Status)
- GenerationRecord(RecordID, AssetID, Timestamp, GeneratedEnergyMWh, AvailabilityPct)
- LoadRecord(LoadID, ZoneID, Timestamp, DemandMW, DemandType)
- PeakEvent(PeakID, ZoneID, StartTime, EndTime, PeakMW, Severity)
- Forecast(ForecastID, Scope, ForecastType, PeriodStart, PeriodEnd, ForecastValueMW, ModelVersion)
- CapacityPlan(PlanID, ZoneID, PlanningHorizon, RecommendedCapacityMW, Notes)

  - SustainabilityMetric(MetricID, Period, RenewableSharePct, EmissionsAvoidedTons, GeneratedDate)
  - ESGReport(ReportID, ReportingStandard, Period, GeneratedDate, Status)
  - ThresholdRule(RuleID, Scope, ThresholdValue, Comparison)
  - Alert(AlertID, RuleID, TriggeredAt, ActualValue, Severity, Status)

7. User Interface Design

- **Grid Analyst Dashboard**: Real-time load vs generation, grid zone KPIs, alerts.
- **Renewable Asset Dashboard**: Generation trends, availability, asset health.
- **Planner Dashboard**: Demand/generation forecasts, capacity plans.
- **ESG Dashboard**: Renewable share, emissions avoided, ESG reports.
- **Admin Dashboard**: Configure zones, assets, thresholds, roles, reports.

8. Non-Functional Requirements

- **Performance**: Handle **high-frequency time-series reads** with up to **50,000 concurrent users** across analytics dashboards.
- **Security**: Encrypted data, RBAC, audit trails.
- **Scalability**: Support large grids, high data volumes, horizontal service scaling.
- **Availability**: Target **99.9% uptime**.
- **Maintainability**: Modular services, API versioning, automated schema migrations.
- **Observability**: Centralized logging and metrics for generation, load, and alerting workflows.

9. Assumptions & Constraints

- Focus on analytics and planning; no real-time SCADA/AMI integration in Phase 1.
- Forecasts are advisory only; no automated grid control.
- All alerts and notifications are in-app only.
- Implementable using standard Java Spring Boot, Angular/React, and relational DB tooling.